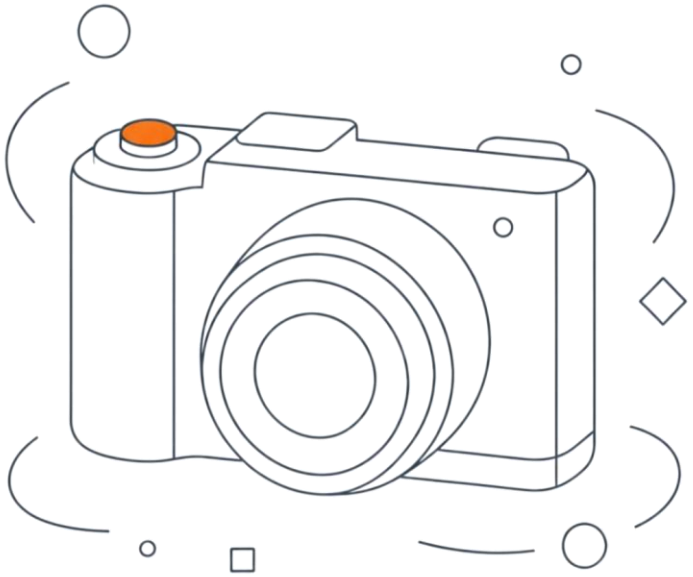
Digital image conversions in Python

From RGB color to grayscale intensity to a binary image.



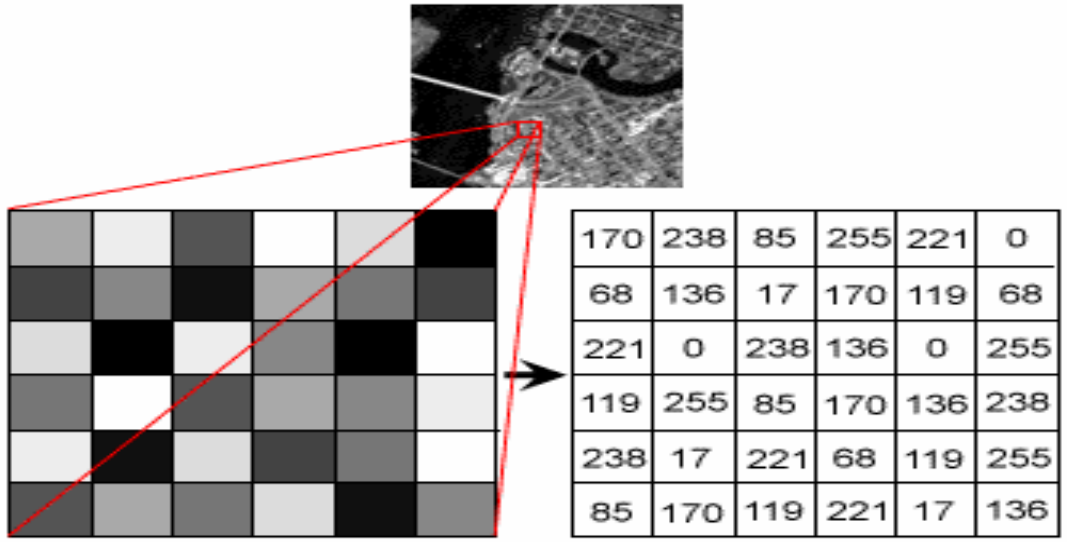
Representation

A digital image is a grid of pixels

Each pixel holds values at one location. Width counts columns; height counts rows. Channels describe what is stored at each location.

40	127	128	210
0	64	192	255
32	96	160	224

Illustrative grayscale grid · 4 columns × 3 rows · 12 pixels.



Zooming in reveals the individual picture elements.

`PNG` and `JPEG` are file formats. `RGB`, `grayscale` and `binary` describe types of digital image.

RGB stores three color channels

One pixel combines red, green and blue values. Pillow RGB mode uses three 8-bit channels, each ranging from 0 to 255: 24 bits per pixel.

R

200

Red channel

G

100

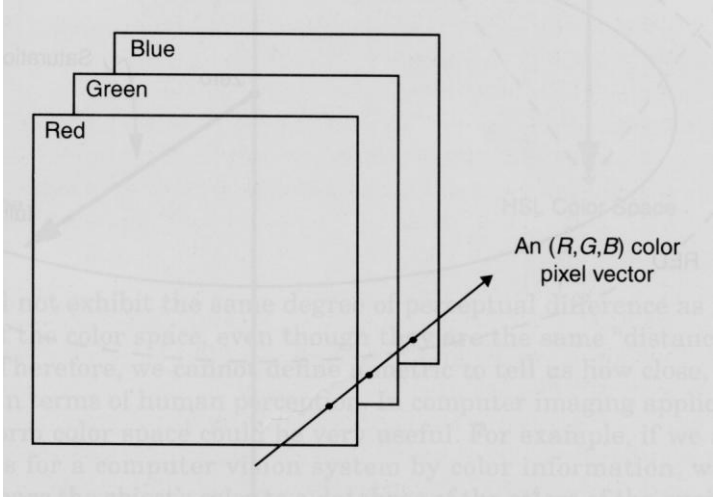
Green channel

B

50

Blue channel

Illustrative channel values. The plates identify the channel colors; the numbers specify their strengths.





Grayscale stores one intensity channel

Grayscale removes color while retaining intensity differences. Pillow L mode uses one 8-bit channel: 0 is black and 255 is white.



Mode	Values per pixel	Information carried
RGB	Three channel values	Color and intensity differences
Grayscale	One intensity value	Intensity differences, without color



Different RGB colors can map to the same gray value. Their original channel values cannot be recovered from grayscale alone.

Binary images contain two pixel states

Every pixel is black or white. A mask uses those states to mark membership, such as foreground versus background in segmentation.

View	Two states
Logical values	0 / 1
Mapping values	0 / 255

Illustrative bilevel mask

0	0	255	255	
0	255	255	0	
0	255	255	0	
0	0	0	0	

No intermediate grays



Convert RGB to grayscale

- Use average of the three channels

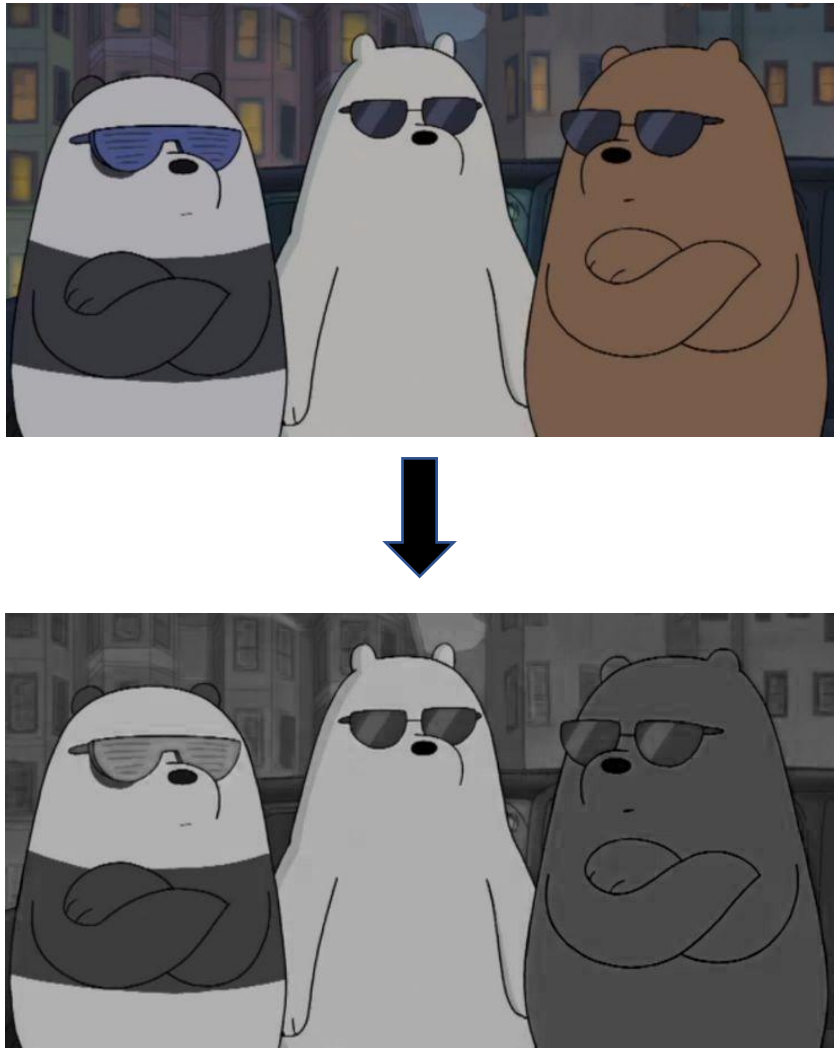
  L=(R+G+B)/3

- Combines the channels using weights.

  *L=0.2*R + 0.3*G + 0.5*B*

  “The sum of the ratios must = 1”

- Use single color channel



Convert grayscale to binary

Choose a boundary `T`. Gray values below it become black; values at or above it become white.

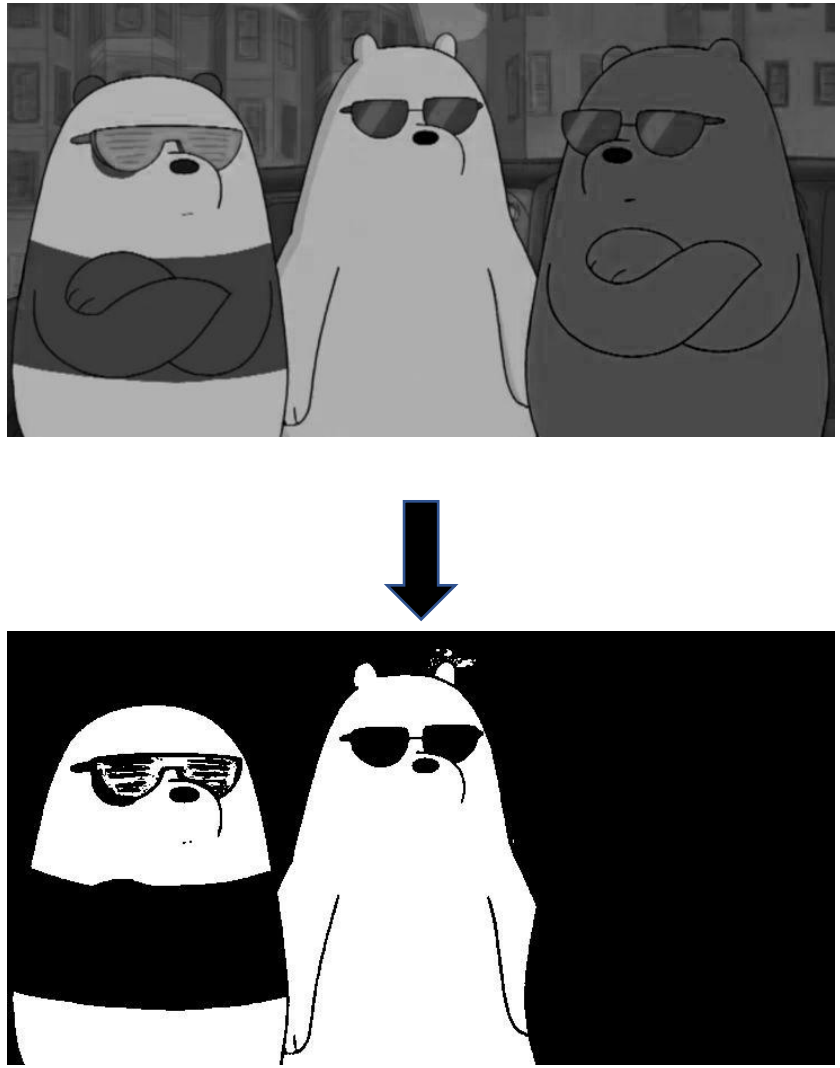
Grayscale input

40	127	128	210

↓ Apply T = 128

0	0	255	255

Binary output





Choose a threshold that fits the task

Mode	Retains	Useful for
RGB	Color and intensity	Color-based analysis
Intenity	Intensity differences	Intensity-based analysis
Binary	A two-state decision	Masks and segmentation